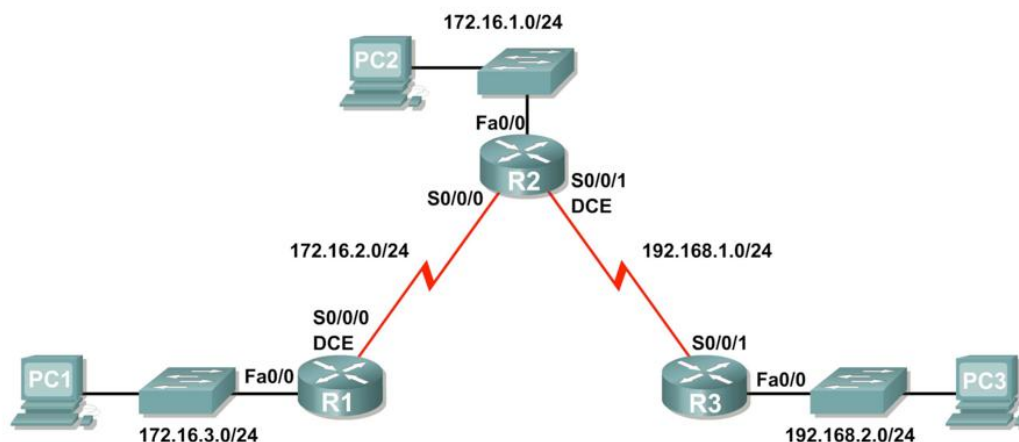


Guião 5 – Encaminhamento estático

Objectivos:

- Endereçamento IP
- Configuração de suporte de rede em terminais linux e em equipamentos de rede cisco
- Configuração e consulta de rotas estáticas

1. Monte a rede de acordo com a ilustração da figura 1.



2. Efetue a configuração dos equipamentos de rede de acordo com os endereços definidos na tabela 1.

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	Fa0/0	172.16.3.1	255.255.255.0	N/A
	S0/0/0	172.16.2.1	255.255.255.0	N/A
R2	Fa0/0	172.16.1.1	255.255.255.0	N/A
	S0/0/0	172.16.2.2	255.255.255.0	N/A
	S0/0/1	192.168.1.2	255.255.255.0	N/A
R3	FA0/0	192.168.2.1	255.255.255.0	N/A
	S0/0/1	192.168.1.1	255.255.255.0	N/A
PC1	NIC	172.16.3.10	255.255.255.0	172.16.3.1
PC2	NIC	172.16.1.10	255.255.255.0	172.16.1.1
PC3	NIC	192.168.2.10	255.255.255.0	192.168.2.1

3. Teste a conectividade do PC para as várias interfaces de rede das máquinas. O que pode concluir?

4. Verifique as rotas conhecidas pelo equipamento de rede.

O que pode verificar da informação apresentada?

5. Configure nos routers o suporte de encaminhamento e verifique o funcionamento da rede.
6. No router 3 configure a interface FastEthernet 0/1 para ter o endereço 10.0.0.1 com a máscara 255.0.0.0, de forma que sirva como porta de acesso à Internet.
7. O acesso à Internet será implementado através da interface FastEthernet 0/1 de R3 que se ligará ao router do ISP com o endereço 10.0.0.254. Configure através do comando ***ip route 0.0.0.0 0.0.0.0 ip_próximo_nó*** os routers de forma que possam permitir o acesso à Internet a toda a rede.
8. Finalize o relatório e submeta-o na plataforma elearning.

Routers GNS3:

Slots:

-> GT96100_FE (2x FastEthernet)

WIC:

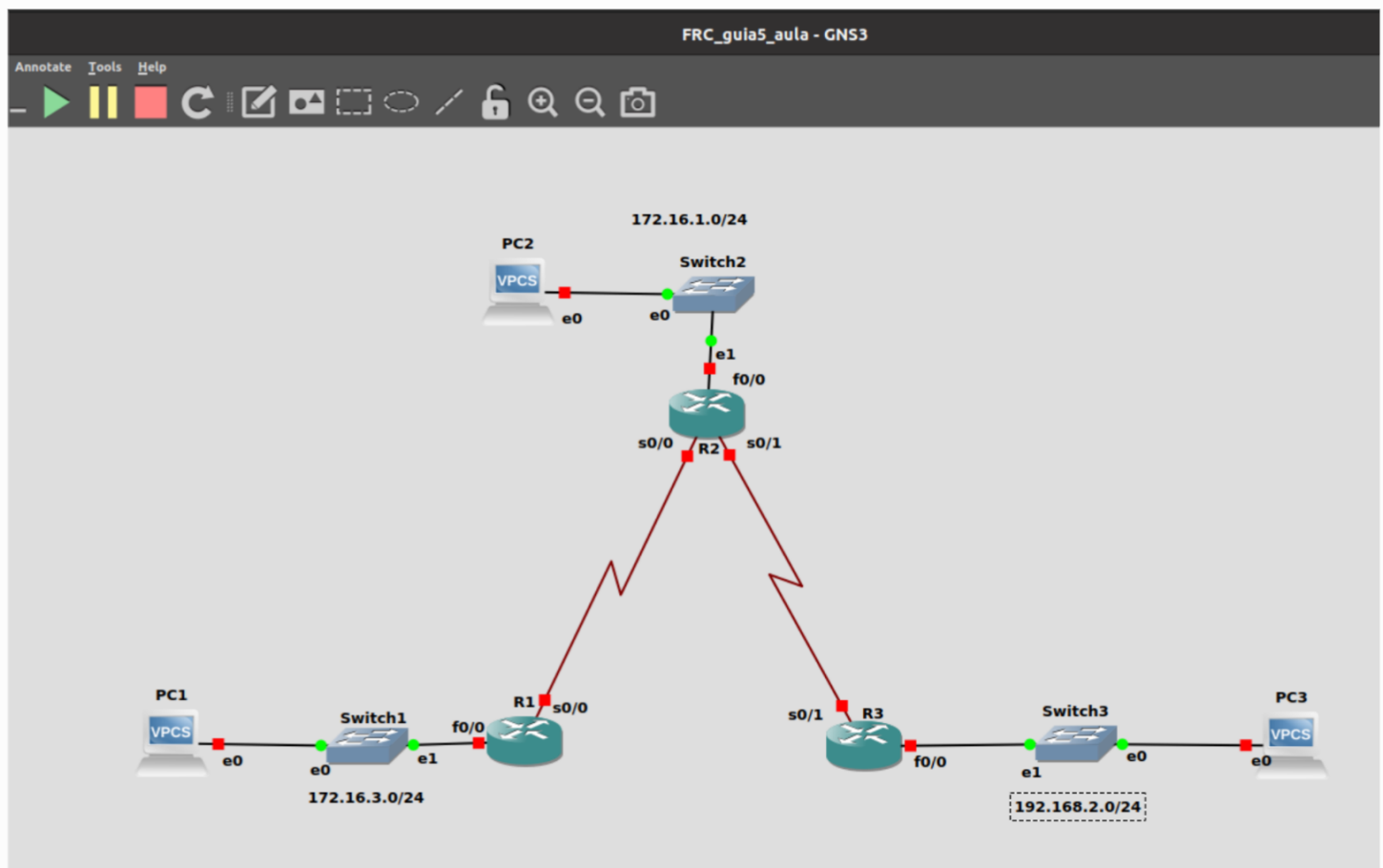
Wic 0: WIC-2T (2x serial s0/0 e s0/1)

Wic 1: WIC-2T (2x serial s1/0 e s1/1)

Wic 2: WIC-2T (2x serial s2/0 e s2/1)

Guia 5:

1.



2.

```
PC1> ip 172.16.3.10 255.255.255.0 172.16.3.1
Checking for duplicate address...
PC1 : 172.16.3.10 255.255.255.0 gateway 172.16.3.1
```

```
PC2> ip 172.16.1.10 255.255.255.0 172.16.1.1
Checking for duplicate address...
PC2 : 172.16.1.10 255.255.255.0 gateway 172.16.1.1
```

```
PC3> ip 192.168.2.10 255.255.255.0 192.168.2.1
Checking for duplicate address...
PC3 : 192.168.2.10 255.255.255.0 gateway 192.168.2.1
```

```
R1(config)#int f0/0
R1(config-if)#ip add
R1(config-if)#ip address 172.16.3.1 255.255.255.0
R1(config-if)#no shut
```

```
R1(config-if)#int s0/0
R1(config-if)#ip add
R1(config-if)#ip address 172.16.2.1 255.255.255.0
R1(config-if)#no shut
R1(config-if)#do sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	172.16.3.1	YES	manual	up	up
Serial0/0	172.16.2.1	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial0/1	unassigned	YES	unset	administratively down	down
Serial0/2	unassigned	YES	unset	administratively down	down
Serial0/3	unassigned	YES	unset	administratively down	down
Serial0/4	unassigned	YES	unset	administratively down	down
Serial0/5	unassigned	YES	unset	administratively down	down
FastEthernet1/0	unassigned	YES	unset	administratively down	down
FastEthernet2/0	unassigned	YES	unset	administratively down	down

```
R2(config)#int f0/0
R2(config-if)#ip add
R2(config-if)#ip address 172.16.1.1 255.255.255.0
R2(config-if)#no shut
```

```
R2(config-if)#int s0/0
R2(config-if)#ip add
R2(config-if)#ip address 172.16.2.2 255.255.255.0
R2(config-if)#no shut
```

```
R2(config-if)#int s0/1
R2(config-if)#ip add
R2(config-if)#ip address
*Mar  1 00:01:52.215: %LINEPROTO-5-UPDOWN: Line protocol o
d state to down
R2(config-if)#ip address 192.168.1.2 255.255.255.0
R2(config-if)#no shut
```

```
R2#sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	172.16.1.1	YES	manual	up	up
Serial0/0	172.16.2.2	YES	manual	up	down
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial0/1	192.168.1.2	YES	manual	up	up
Serial0/2	unassigned	YES	unset	administratively down	down
Serial0/3	unassigned	YES	unset	administratively down	down
Serial0/4	unassigned	YES	unset	administratively down	down
Serial0/5	unassigned	YES	unset	administratively down	down
FastEthernet1/0	unassigned	YES	unset	administratively down	down
FastEthernet2/0	unassigned	YES	unset	administratively down	down


```
R3(config)#int f0/0
R3(config-if)#ip add
R3(config-if)#ip address 192.168.2.1 255.255.255.0
R3(config-if)#no shut
```

```
R3(config-if)#int s0/1
R3(config-if)#ip add
R3(config-if)#ip address 192.168.1.1 255.255.255.0
R3(config-if)#no shut
```

```
R3#sh ip int br
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	192.168.2.1	YES	manual	up	up
Serial0/0	unassigned	YES	unset	administratively down	down
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Serial0/1	192.168.1.1	YES	manual	up	down
Serial0/2	unassigned	YES	unset	administratively down	down
Serial0/3	unassigned	YES	unset	administratively down	down
Serial0/4	unassigned	YES	unset	administratively down	down
Serial0/5	unassigned	YES	unset	administratively down	down
FastEthernet1/0	unassigned	YES	unset	administratively down	down
FastEthernet2/0	unassigned	YES	unset	administratively down	down

3. e 4.:

```
PC1> ping 172.16.1.10
```

```
*172.16.3.1 icmp_seq=1 ttl=255 time=819.407 ms (ICMP type:3, code:1, Destination host unreachable)
*172.16.3.1 icmp_seq=2 ttl=255 time=12.156 ms (ICMP type:3, code:1, Destination host unreachable)
*172.16.3.1 icmp_seq=3 ttl=255 time=49.281 ms (ICMP type:3, code:1, Destination host unreachable)
*172.16.3.1 icmp_seq=4 ttl=255 time=12.020 ms (ICMP type:3, code:1, Destination host unreachable)
*172.16.3.1 icmp_seq=5 ttl=255 time=9.034 ms (ICMP type:3, code:1, Destination host unreachable)
```

```
PC1> ping 192.168.2.10
```

```
*172.16.3.1 icmp_seq=1 ttl=255 time=10.103 ms (ICMP type:3, code:1, Destination host unreachable)
*172.16.3.1 icmp_seq=2 ttl=255 time=48.683 ms (ICMP type:3, code:1, Destination host unreachable)
*172.16.3.1 icmp_seq=3 ttl=255 time=14.075 ms (ICMP type:3, code:1, Destination host unreachable)
*172.16.3.1 icmp_seq=4 ttl=255 time=13.785 ms (ICMP type:3, code:1, Destination host unreachable)
*172.16.3.1 icmp_seq=5 ttl=255 time=7.365 ms (ICMP type:3, code:1, Destination host unreachable)
```

```
R1#sh ip route
```

Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

```
172.16.0.0/24 is subnetted, 2 subnets
C      172.16.2.0 is directly connected, Serial0/0
C      172.16.3.0 is directly connected, FastEthernet0/0
```

```
R2#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route
```

Gateway of last resort is not set

```
      172.16.0.0/24 is subnetted, 2 subnets
C      172.16.1.0 is directly connected, FastEthernet0/0
C      172.16.2.0 is directly connected, Serial0/0
C      192.168.1.0/24 is directly connected, Serial0/1
```

```
R3#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route
```

Gateway of last resort is not set

```
C      192.168.1.0/24 is directly connected, Serial0/1
C      192.168.2.0/24 is directly connected, FastEthernet0/0
```

Posso concluir que os PCs não conseguem comunicar entre si. Isto acontece porque os routers não têm rotas para as redes que não estão diretamente conectadas a si (como demonstrado pelas routing tables dos routers). O R1 não conhece as redes 172.16.1.0/24, 192.168.1.0/24 e 192.168.2.0/24. O R2 não conhece as redes 172.16.3.0/24, 192.168.2.0/24. O R3 não conhece as redes 172.16.1.0/24, 172.16.2.0/24 e 172.16.3.0/24.

5.

R1:

```
ip route 172.16.1.0 255.255.255.0 172.16.2.2
ip route 192.168.2.0 255.255.255.0 172.16.2.2
ip route 192.168.1.0 255.255.255.0 172.16.2.2
```

R2:

```
ip route 172.16.3.0 255.255.255.0 172.16.2.1
ip route 192.168.2.0 255.255.255.0 192.168.1.1
```

R3:

ip route 172.16.1.0 255.255.255.0 192.168.1.2

ip route 172.16.2.0 255.255.255.0 192.168.1.2

ip route 172.16.3.0 255.255.255.0 192.168.1.2

```
R1(config)#ip route 172.16.1.0 255.255.255.0 172.16.2.2
R1(config)#ip route 192.168.2.0 255.255.255.0 172.16.2.2
R1(config)#ip route 192.168.1.0 255.255.255.0 172.16.2.2
R1(config)#end
R1#
*Mar  1 00:02:40.559: %SYS-5-CONFIG_I: Configured from console by console
R1#write
Building configuration...
[OK]
R1#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/24 is subnetted, 3 subnets
S       172.16.1.0 [1/0] via 172.16.2.2
C       172.16.2.0 is directly connected, Serial0/0
C       172.16.3.0 is directly connected, FastEthernet0/0
S       192.168.1.0/24 [1/0] via 172.16.2.2
S       192.168.2.0/24 [1/0] via 172.16.2.2
```

```
R2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#ip route 172.16.3.0 255.255.255.0 172.16.2.1
R2(config)#ip route 192.168.2.0 255.255.255.0 192.168.1.1
R2(config)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/24 is subnetted, 3 subnets
C       172.16.1.0 is directly connected, FastEthernet0/0
C       172.16.2.0 is directly connected, Serial0/0
S       172.16.3.0 [1/0] via 172.16.2.1
C       192.168.1.0/24 is directly connected, Serial0/1
S       192.168.2.0/24 [1/0] via 192.168.1.1
```



```

R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
R3(config)#ip route 172.16.1.0 255.255.255.0 192.168.1.2
R3(config)#ip route 172.16.2.0 255.255.255.0 192.168.1.2
R3(config)#ip route 172.16.3.0 255.255.255.0 192.168.1.2
R3(config)#end
R3#
*Mar  1 01:01:10.191: %SYS-5-CONFIG_I: Configured from console by console
R3#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/24 is subnetted, 3 subnets
S       172.16.1.0 [1/0] via 192.168.1.2
S       172.16.2.0 [1/0] via 192.168.1.2
S       172.16.3.0 [1/0] via 192.168.1.2
C       192.168.1.0/24 is directly connected, Serial0/1
C       192.168.2.0/24 is directly connected, FastEthernet0/0

```

E só depois, fazer pings:

```

PC1> ping 172.16.1.10

84 bytes from 172.16.1.10 icmp_seq=1 ttl=62 time=39.584 ms
84 bytes from 172.16.1.10 icmp_seq=2 ttl=62 time=23.368 ms
84 bytes from 172.16.1.10 icmp_seq=3 ttl=62 time=15.047 ms
172.16.1.10 icmp_seq=4 timeout
84 bytes from 172.16.1.10 icmp_seq=5 ttl=62 time=21.610 ms

PC1> ping 192.168.2.10

192.168.2.10 icmp_seq=1 timeout
84 bytes from 192.168.2.10 icmp_seq=2 ttl=61 time=15.727 ms
84 bytes from 192.168.2.10 icmp_seq=3 ttl=61 time=21.511 ms
84 bytes from 192.168.2.10 icmp_seq=4 ttl=61 time=591.114 ms
84 bytes from 192.168.2.10 icmp_seq=5 ttl=61 time=480.987 ms

```

Já existe conectividade entre os três PCs

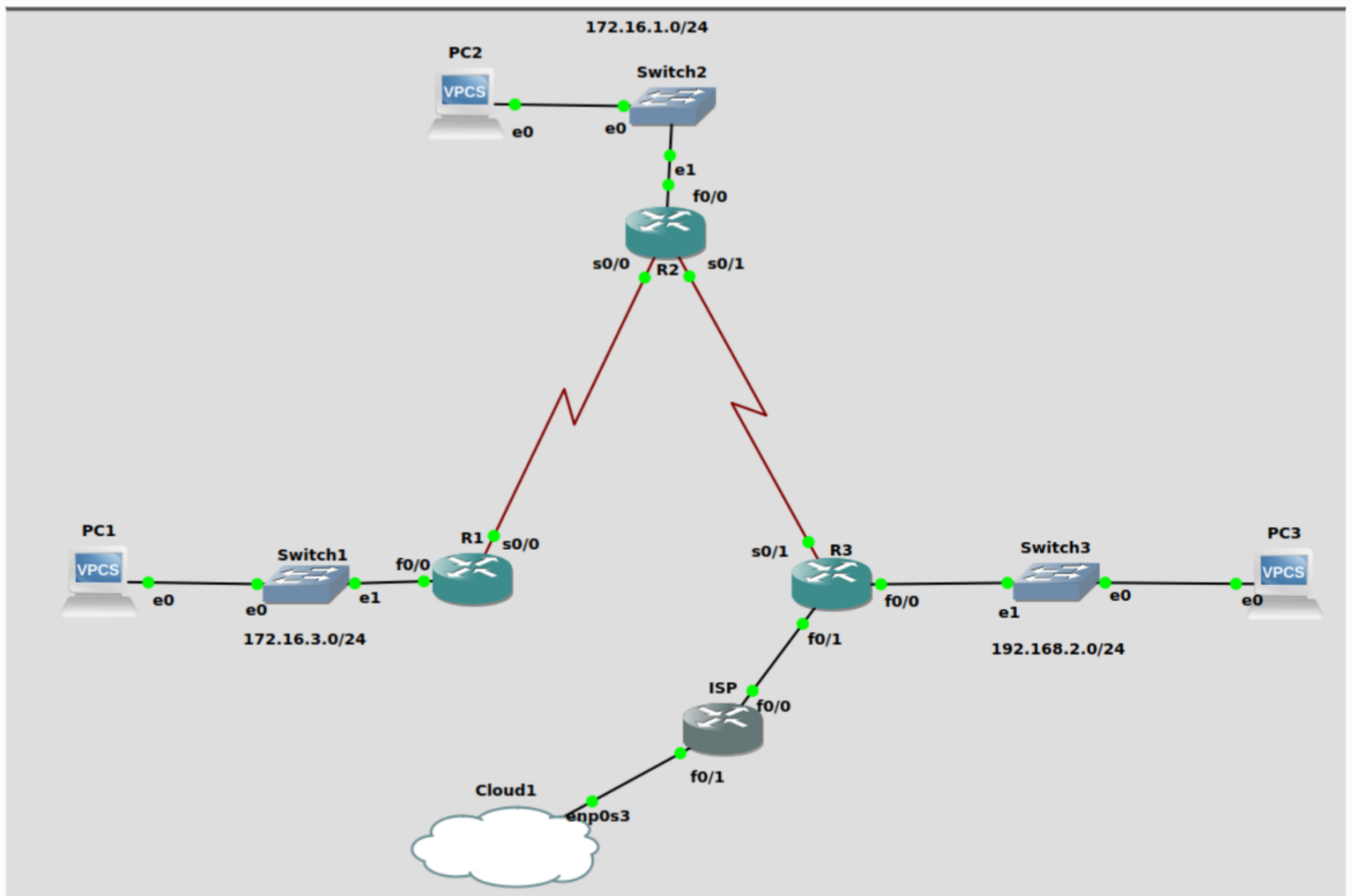
6.

```

R3(config)#int f0/1
R3(config-if)#
R3(config-if)#ip add
R3(config-if)#ip address 10.0.0.1 255.0.0.0
R3(config-if)#no shut
R3(config-if)#
*Mar  1 00:13:18.927: %LINK-3-UPDOWN: Interface FastEthernet0/1, changed state to up
*Mar  1 00:13:19.927: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

```

7.



Router ISP:

```
ISP(config)#int f0/0
ISP(config-if)#ip address 10.0.0.254 255.0.0.0

ISP(config-if)#no shut
ISP(config-if)#
*Mar 1 00:03:25.543: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar 1 00:03:26.543: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, change
state to up

ISP(config)#ip route 0.0.0.0 0.0.0.0 f0/1
```

```
ip route 172.16.1.0 255.255.255.0 10.0.0.1
ip route 172.16.2.0 255.255.255.0 10.0.0.1
ip route 172.16.3.0 255.255.255.0 10.0.0.1
ip route 192.168.1.0 255.255.255.0 10.0.0.1
ip route 192.168.2.0 255.255.255.0 10.0.0.1
```

```

ISP#conf t
Enter configuration commands, one per line. End with CNTL/Z.
ISP(config)#ip route 172.16.1.0 255.255.255.0 10.0.0.1
ISP(config)#
ISP(config)#ip route 172.16.2.0 255.255.255.0 10.0.0.1
ISP(config)#ip route 172.16.3.0 255.255.255.0 10.0.0.1
ISP(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
ISP(config)#
ISP(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.1
ISP(config)#end
ISP#
*Mar 1 00:01:49.243: %SYS-5-CONFIG_I: Configured from console by console
ISP#sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

    172.16.0.0/24 is subnetted, 3 subnets
S       172.16.1.0 [1/0] via 10.0.0.1
S       172.16.2.0 [1/0] via 10.0.0.1
S       172.16.3.0 [1/0] via 10.0.0.1
C    10.0.0.0/8 is directly connected, FastEthernet0/0
S    192.168.1.0/24 [1/0] via 10.0.0.1
S    192.168.2.0/24 [1/0] via 10.0.0.1

```

R3:

ip route 0.0.0.0 0.0.0.0 10.0.0.254

do sh ip route

```

R3(config)#ip route 0.0.0.0 0.0.0.0 10.0.0.254
R3(config)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route

Gateway of last resort is 10.0.0.254 to network 0.0.0.0

    172.16.0.0/24 is subnetted, 3 subnets
S       172.16.1.0 [1/0] via 192.168.1.2
S       172.16.2.0 [1/0] via 192.168.1.2
S       172.16.3.0 [1/0] via 192.168.1.2
C    10.0.0.0/8 is directly connected, FastEthernet0/1
C    192.168.1.0/24 is directly connected, Serial0/1
C    192.168.2.0/24 is directly connected, FastEthernet0/0
S*    0.0.0.0/0 [1/0] via 10.0.0.254

```

R2:

ip route 0.0.0.0 0.0.0.0 192.168.1.1

```
R2(config)#ip route 0.0.0.0 0.0.0.0 192.168.1.1
R2(config)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 192.168.1.1 to network 0.0.0.0

    172.16.0.0/24 is subnetted, 3 subnets
C       172.16.1.0 is directly connected, FastEthernet0/0
C       172.16.2.0 is directly connected, Serial0/0
S       172.16.3.0 [1/0] via 172.16.2.1
C       192.168.1.0/24 is directly connected, Serial0/1
S       192.168.2.0/24 [1/0] via 192.168.1.1
S*      0.0.0.0/0 [1/0] via 192.168.1.1
```

R1:

ip route 0.0.0.0 0.0.0.0 172.16.2.2

```
R1(config)#ip route 0.0.0.0 0.0.0.0 172.16.2.2
R1(config)#
R1(config)#do sh ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 172.16.2.2 to network 0.0.0.0

    172.16.0.0/24 is subnetted, 3 subnets
S       172.16.1.0 [1/0] via 172.16.2.2
C       172.16.2.0 is directly connected, Serial0/0
C       172.16.3.0 is directly connected, FastEthernet0/0
S       192.168.1.0/24 [1/0] via 172.16.2.2
S       192.168.2.0/24 [1/0] via 172.16.2.2
S*      0.0.0.0/0 [1/0] via 172.16.2.2
```

PC1> ping 10.0.0.1

```
10.0.0.1 icmp_seq=1 timeout
84 bytes from 10.0.0.1 icmp_seq=2 ttl=253 time=21.321 ms
84 bytes from 10.0.0.1 icmp_seq=3 ttl=253 time=12.207 ms
10.0.0.1 icmp_seq=4 timeout
84 bytes from 10.0.0.1 icmp_seq=5 ttl=253 time=1001.212 ms
```

PC1> ping 10.0.0.254

```
10.0.0.254 icmp_seq=1 timeout
84 bytes from 10.0.0.254 icmp_seq=2 ttl=252 time=503.470 ms
84 bytes from 10.0.0.254 icmp_seq=3 ttl=252 time=282.702 ms
10.0.0.254 icmp_seq=4 timeout
84 bytes from 10.0.0.254 icmp_seq=5 ttl=252 time=49.923 ms
```



```
PC1> ping 159.12.0.3

159.12.0.3 icmp_seq=1 timeout
159.12.0.3 icmp_seq=2 timeout
*10.0.0.254 icmp_seq=3 ttl=252 time=81.450 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.0.254 icmp_seq=4 ttl=252 time=32.186 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.0.254 icmp_seq=5 ttl=252 time=21.318 ms (ICMP type:3, code:1, Destination host unreachable)
```

```
PC2> ping 162.2.222.33

162.2.222.33 icmp_seq=1 timeout
*10.0.0.254 icmp_seq=2 ttl=253 time=33.981 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.0.254 icmp_seq=3 ttl=253 time=23.201 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.0.254 icmp_seq=4 ttl=253 time=25.376 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.0.254 icmp_seq=5 ttl=253 time=30.211 ms (ICMP type:3, code:1, Destination host unreachable)
```

```
PC3> ping 13.55.98.1

13.55.98.1 icmp_seq=1 timeout
*10.0.0.254 icmp_seq=2 ttl=254 time=32.770 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.0.254 icmp_seq=3 ttl=254 time=414.768 ms (ICMP type:3, code:1, Destination host unreachable)
)
*10.0.0.254 icmp_seq=4 ttl=254 time=37.754 ms (ICMP type:3, code:1, Destination host unreachable)
*10.0.0.254 icmp_seq=5 ttl=254 time=24.596 ms (ICMP type:3, code:1, Destination host unreachable)
```

(pings para a "Internet" não funcionaram porque os destinos não existem, mas chegaram lá.)